

# Mukund Yedunuthala

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## SUMMARY

Computational Materials Scientist with a background in applied mathematics, continuum mechanics, and high-performance scientific computing. Experienced in numerical modelling, PDE discretisation, nonlinear finite-element methods, numerical linear algebra, and scientific programming in Python, MATLAB, and C++ for validated simulation workflows.

## SKILLS

**Programming Languages:** Python, MATLAB, C++, Fortran, Rust

**Numerical Methods & Scientific Computing:** PDE discretisation, finite element methods, numerical linear algebra, Krylov methods (CG, PCG, GMRES), Additive Schwarz preconditioning, NumPy, SciPy, PETSc, MATLAB

**Distributed, Parallel & GPU Computing:** MPI, OpenMP, CUDA, Numba, PBS, SLURM

**Software Development:** Linux, Git, PyTest, CMake, Docker, GoogleTest, Sphinx, Doxygen

**Physical Modelling & Simulation:** Abaqus, COMSOL Multiphysics, LS-DYNA, ANSYS Workbench, continuum mechanics, structural mechanics, heat transfer, diffusion, radiation

**Machine Learning:** PyTorch, TensorFlow, scikit-learn, model optimisation

**Languages:** German (B2), English (C1)

## EDUCATION

**Master of Science, Computational Materials Science (Engl.)**

Oct 2019 – Sep 2024

TU Bergakademie Freiberg

Freiberg, Germany

Interdisciplinary programme combining applied mathematics, continuum mechanics, and high-performance computing — the curriculum demanded both mathematical rigour and working, validated implementations.

- Relevant Modules: Numerical Analysis of Differential Equations, Introduction to Scientific Programming, Introduction to High-Performance Computing, Nonlinear Finite Element Methods, Continuum Mechanics, Plasticity, Machine Learning for Materials Scientists
- Master's Thesis: "Overlapping Schwarz Domain Decomposition Methods in Python with Applications in Structural Mechanics" — developed and evaluated a nonlinear FEM solver using domain-decomposition-based preconditioning, benchmarking Krylov methods against an Additive Schwarz preconditioner (Thesis grade: 2.3).
- Final grade: 2.3. Supervisors: Prof. Dr. Oliver Rheinbach, Dr. Stephan Köhler.

**Bachelor of Engineering, Mechanical Engineering**

Aug 2015 – Jul 2019

Chaitanya Bharathi Institute of Technology

Hyderabad, India

- Selected Coursework: Engineering Mechanics, Strength of Materials, Fluid Mechanics, Thermodynamics, Heat Transfer, Computer-Aided Design and Manufacturing, Machine Design, Dynamics of Machines
- Bachelor's Thesis: "Design and Analysis of Air Bearing Frame for a Hyperloop Transportation System" (Supervisor: Dr. P. Ravinder Reddy)
- Final grade: 8.6/10

## EXPERIENCE

**Technical Intern — AI Model Optimisation**

Oct 2024 – Apr 2025

Aientasar Software Solutions

Hyderabad, India

- Analysed numerical performance and model evaluation pipelines for computer-vision anomaly detection using Python and TensorFlow.
- Owned data preprocessing, validation, and robustness analysis for camera-feed image data.
- Improved model deployment readiness in collaboration with the engineering team.

**Research Assistant (HiWi) & Master's Thesis**

Aug 2023 – Sep 2024

Institute of Numerical Mathematics and Optimization, TU Bergakademie Freiberg

Freiberg, Germany

- Formulated and implemented a nonlinear FEM solver in Python for a von Mises plasticity model under small-strain assumptions.
- Applied numerical methods for discretised PDE systems, including Newton-Raphson, Krylov solvers (CG, PCG, GMRES), and UMFPAK.
- Implemented overlapping Additive Schwarz preconditioning and benchmarked the solver against conventional Newton and Abaqus reference simulations.

- Maintained code reproducibility and modularity through version control (Git) and structured testing (PyTest unit and integration tests).

### **Summer Intern — PRACE Summer of HPC 2021**

Jun 2021 – Aug 2021

MdLS – Maison de la Simulation (CEA/CNRS)

France / Remote

- Selected as one of 66 participants for the PRACE Summer of HPC programme.
- Studied an existing parallel FEM solver codebase, written in C/C++ with OpenMP and MPI support, across several use cases and configurations.
- Collaborated with an international team to extend the parallel FEM codebase for heat transfer through diffusion and radiation.
- Received practical training in MPI, OpenMP, GPU programming with CUDA and Numba, and batch scheduling on a computing cluster using SLURM.

## **PROJECTS**

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### **Overlapping Schwarz Domain Decomposition Methods in Python with Applications in Structural Mechanics**

Aug 2023 – Sep 2024

Python (NumPy, SciPy, PyTest, Sphinx), PETSc, Abaqus, LaTeX

Freiberg, Germany

- Built a finite element workflow for nonlinear physical modelling by decomposing structural-mechanics meshes into overlapping subdomains.
- Implemented Newton's method with CG, PCG, GMRES, UMFPACK, and Additive Schwarz preconditioning for the discretised PDE system.
- Achieved an 8x per-step runtime reduction, from 160 s to 20 s, versus a vanilla Newton solver at the same problem scale.
- Validated and cross-compared simulation results with Abaqus reference models for physical correctness.
- [mukundyedunuthala.page/blog/ddm-nfem](https://mukundyedunuthala.page/blog/ddm-nfem)

### **Modelling of Radiative Heat Exchange Using the Finite Element Method**

Nov 2022 – May 2023

Python (NumPy, SciPy, PyTest, Sphinx, Matplotlib), Intel Embree, GMsh

Freiberg, Germany

- Implemented a FEM solver for coupled heat transfer by diffusion and radiation, with an object-oriented wrapper and GMsh mesh importer.
- Built a targeted Python binding for Intel Embree's ray-tracing API to compute geometric view factors between mesh surfaces.
- Profiled and resolved performance bottlenecks using Python's built-in profiling tools.
- Version-controlled the project with Git, implemented unit tests, and documented the codebase using Sphinx and a LaTeX report.
- [github.com/mukund-yedunuthala/radiative-heat-fem](https://github.com/mukund-yedunuthala/radiative-heat-fem)

### **Image Convolution using Message Passing Interface (MPI)**

Oct 2020 – Apr 2021

C++ (MPICH/OpenMPI, GoogleTest), CMake, PBS

Freiberg, Germany

- Implemented sequential and distributed image convolution from scratch in C++ using MPI and configurable numerical kernels.
- Performed a scalability study on an image convolution algorithm with multiple kernels.
- Developed automated, configurable workflows for building, testing, and documenting using CMake.
- Performed the scalability study on the university's computing cluster using the PBS queueing system.
- [github.com/mukund-yedunuthala/hpc-img-convolution](https://github.com/mukund-yedunuthala/hpc-img-convolution)

## **CERTIFICATIONS**

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**PyTorch for Deep Learning Professional Certificate** — DeepLearning.AI (July 2026)

**Fortran for Scientific Computing** — PRACE and Vlaams Supercomputer Centre (March 2026)

**Fundamentals of Accelerated Computing with CUDA Python** — NVIDIA Deep Learning Institute (August 2024)

**Deutsch: Kommunikation in Studium und Beruf** — International University Center (IUZ), TU Bergakademie Freiberg (April 2024)